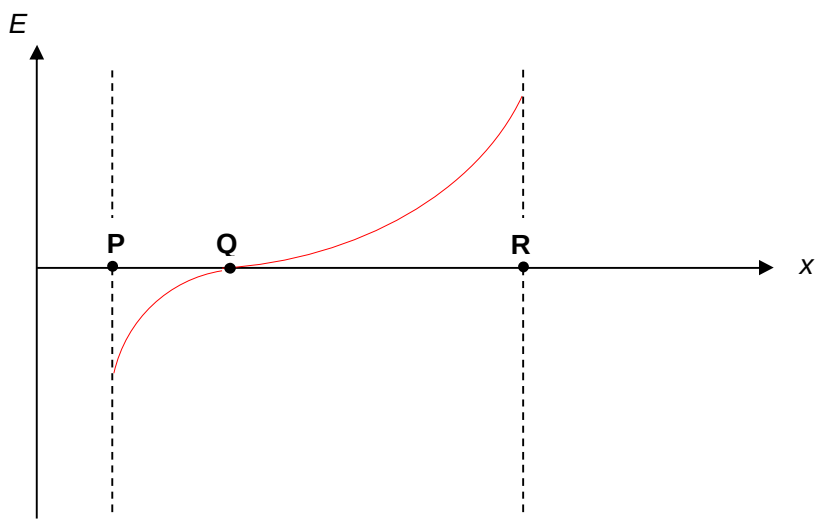


4	(a)	The total <u>electric potential</u> is <u>negative for all points</u> between them. <u>Both</u> conductors A and B are <u>negatively charged</u> .	M1 A1
	(b)	Field <u>strength</u> of the electric field at a point is numerically equal to the (negative) <u>potential gradient</u> at that point At <u>Q (or turning point)</u> , the <u>resultant electric field strength is zero</u> . Since <u>Q (or turning point)</u> is <u>closer to A</u> (or further from B), <u>conductor B</u> has a larger magnitude of charge.	B1 B1 B1
	(c)	work done by external force = change in electric potential energy = $q (V_R - V_P) = (-2.0)(-95.0 + 30.0)$ = + 130 J <i>Allow BOD if substitution is $(2.0)(95.0 - 30.0)$</i>	M1 A1
	(d)	 Fig. 3.2 Correct shape [M1] E_P is zero at point Q, magnitude of E_P smaller than E_R and both touching the dotted boundary lines [A1]	

5	(a)	(i)	The magnetic flux through a plane surface is the product of the	
---	-----	-----	---	--